Retail Demand Forecasting for Intermittent Demand

Comparative Evaluation of Traditional and Specialized Forecasting Methods

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Business Problem

The Challenge

Thousands of Products (SKUs)

Hundreds of Stores

Continuously changing demand

Inventory Decisions

Too much inventory → Higher holding costs

Too little inventory → Stockouts & lost sales

The Real Challenge: Many retail products exhibit intermittent demand

Long zero-demand periods

Irregular purchases

Sudden spikes

Selecting the right forecasting method is critical for balancing inventory cost and product availability.

Project Objective

Develop a forecasting framework for intermittent retail demand.

Objectives

✓ Understand demand characteristics

✓ Determine appropriate forecasting granularity

✓ Perform Exploratory Data Analysis

✓ Compare forecasting models
  - Moving Average
  - Croston
  - SBA
  - TSB

✓ Recommend the most suitable forecasting approach

Dataset

Daily Retail Transactions

↓

3.3 Million Records

↓

200 Stores

↓

50 Products

↓

Sales Volume
(Target)

Available Variables
Calendar Date
Store ID
Product ID
Sales Value
Sales Volume

Data Preparation

Why not Daily?

- Too many zero-demand observations
- High forecasting instability

Why Weekly?

- Reduces sparsity
- Preserves purchasing behavior
- Stable demand patterns

Weekly Dataset

≈ **478K**

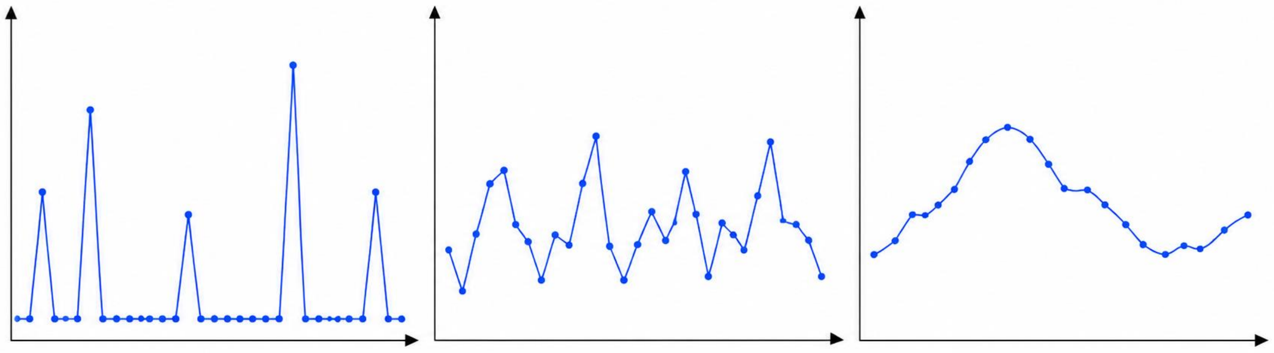
Store–Product–Week observations

70 Weeks

Store–Product–Week forecasting granularity

Why Weekly Aggregation?

Daily	Weekly	Monthly
❌ Too Sparse	✅ Best Balance	❌ Too Smooth



Weekly aggregation transformed noisy daily transactions into a stable forecasting dataset.

How were promotions identified?

Business Problem

- Dataset did not contain a promotion indicator.
- Promotions significantly influence retail demand.
- Promotions needed to be identified before analyzing their impact.

Methodology

Estimated Regular Price

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Calculated Price Drop %

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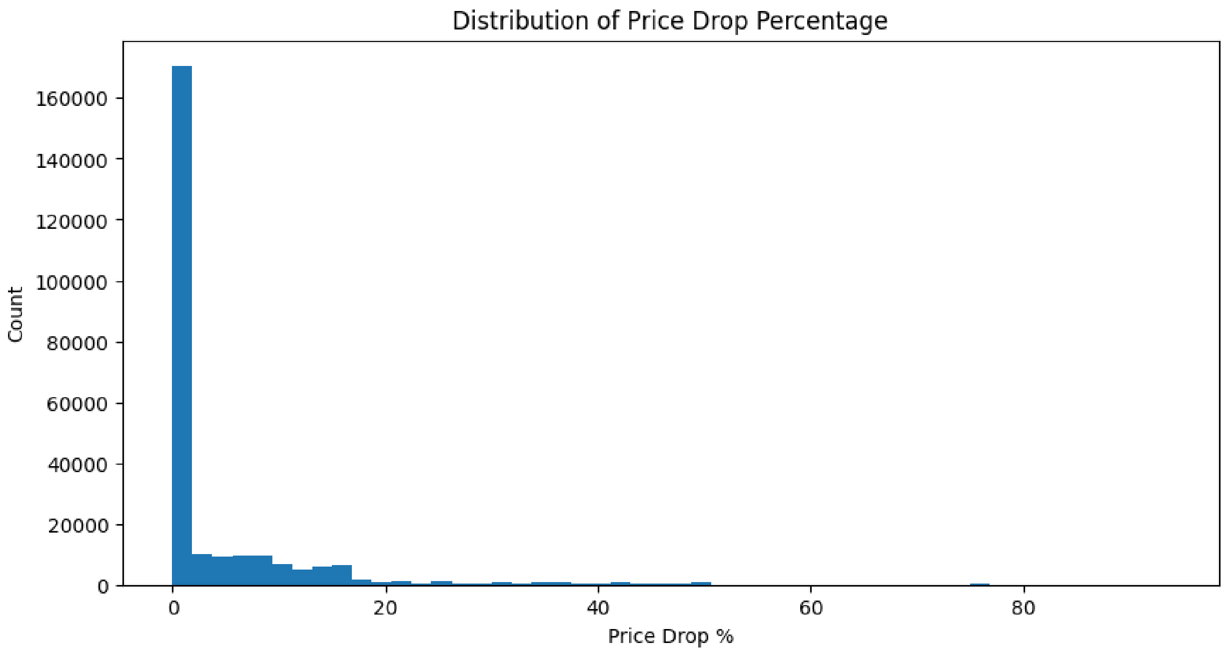
Classified significant temporary price reductions as Promotions

Key Finding

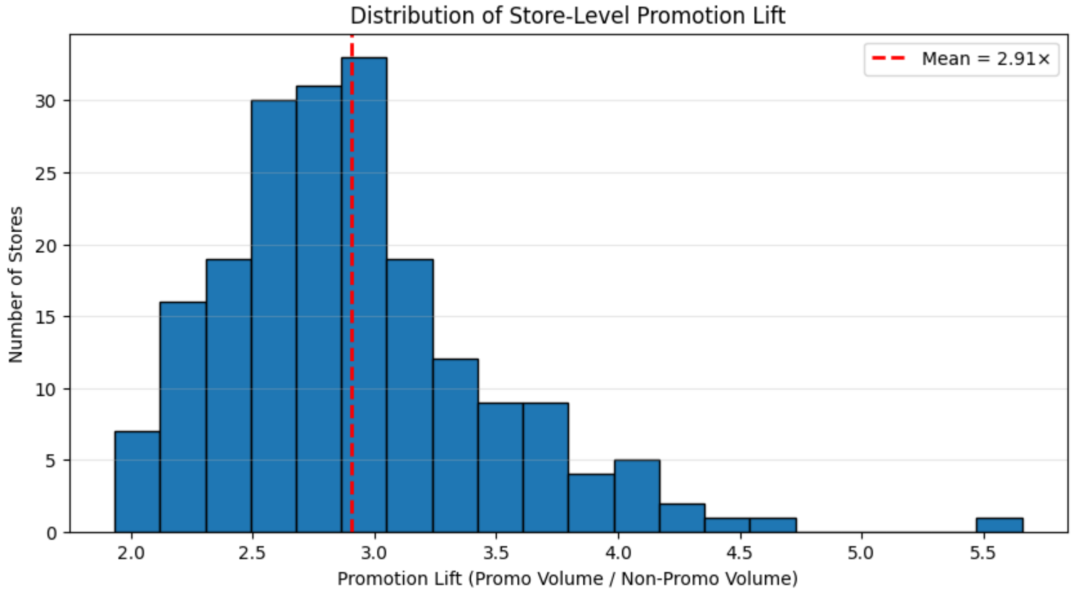
- Most observations had little or no discount.
- Only ***~12%*** of store-product-week observations were promotional.



Promotions are **infrequent but high-impact events**, making them an important explanatory variable rather than the baseline demand pattern.

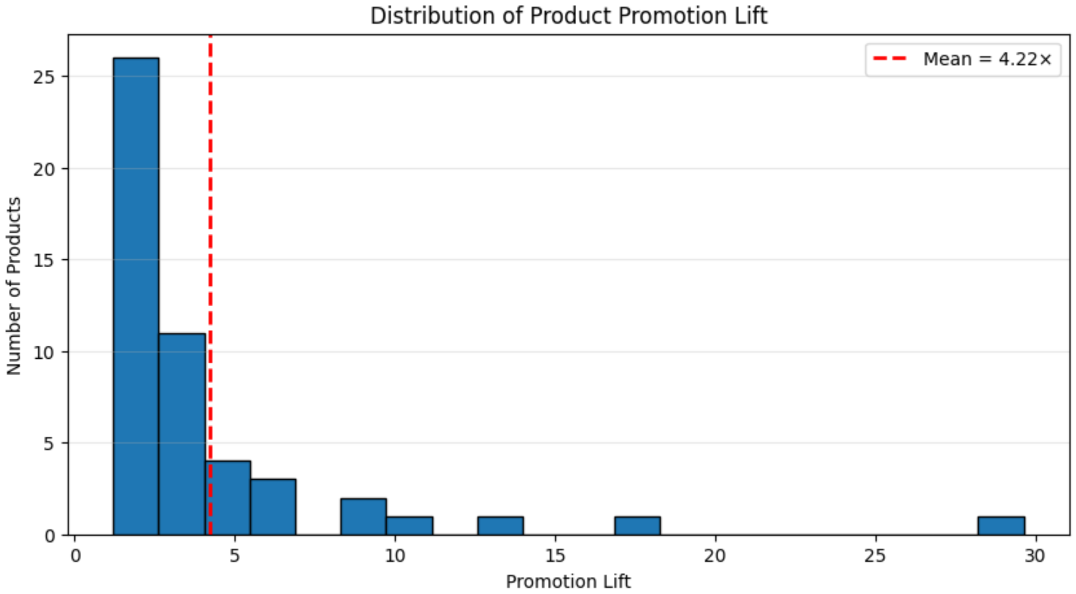
Do Promotions Affect All Stores and Products Equally?

Store-Level Analysis



Purpose
Determine whether promotional effectiveness is consistent across retail locations.
Observation
Promotions increase demand across most stores.
Magnitude of uplift varies significantly.
Business Insight
Customer behavior differs by location.

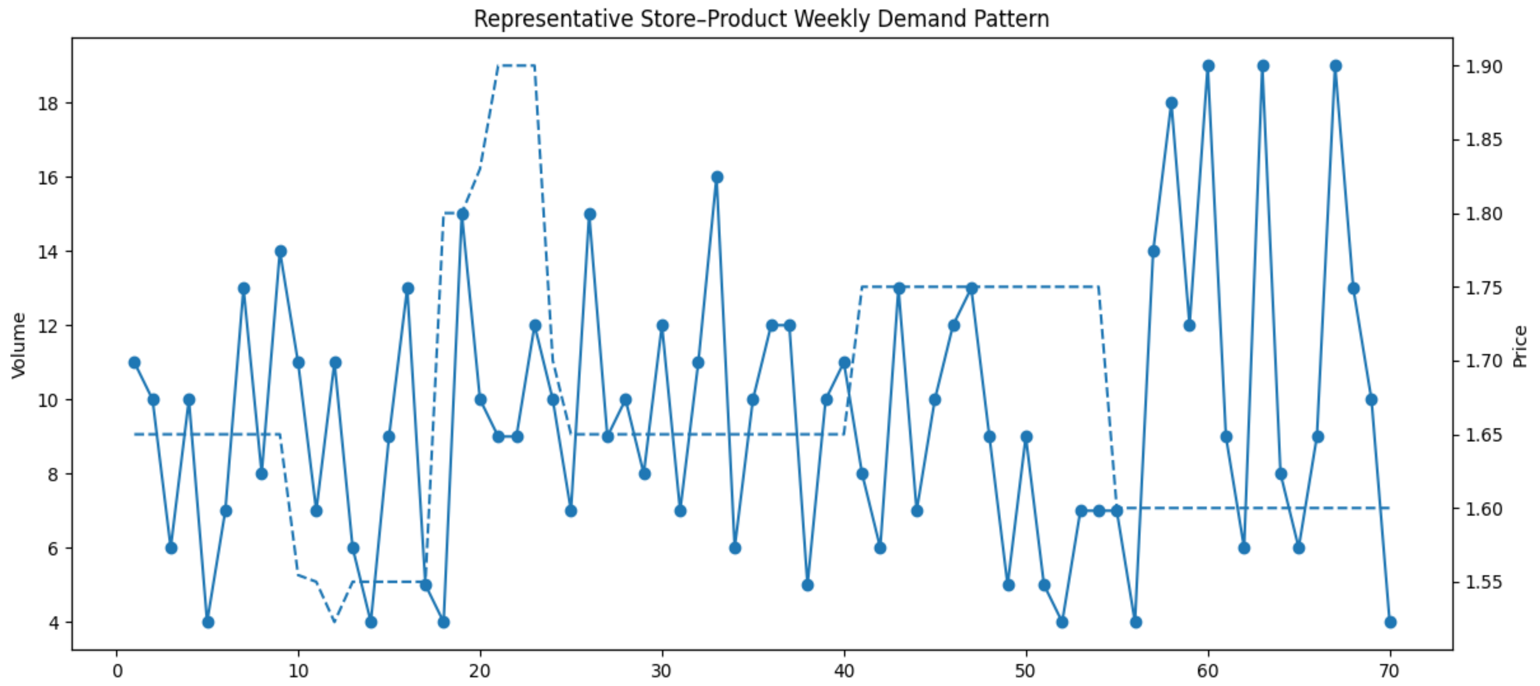
Product-Level Analysis



Purpose
Evaluate whether all products respond similarly to promotions.
Observation
Promotion lift varies widely across products.
Some products experience large demand increases.
Others show relatively small changes.
Business Insight
Products have different price sensitivity and purchasing behavior.

Forecasting should be performed at the Store–Product level while retaining promotion-related features.

Is the Demand Actually Intermittent?



Why analyze an individual demand series?

Aggregate statistics can hide irregular purchasing behavior.

A representative store-product series helps visualize the real forecasting challenge.

Observations

✓ Long periods with zero demand

✓ Irregular demand occurrence

✓ Sudden demand spikes

✓ No consistent trend

Business Insight

Many retail products are purchased **only when customers need them**, rather than on a regular schedule.

This creates sparse and unpredictable demand.

Forecasting Impact

Traditional forecasting methods assume continuous demand.

Specialized intermittent forecasting methods should therefore be evaluated.

What Type of Demand Dominates the Dataset?

ADI (Average Demand Interval)

Business Perspective

How frequently customers purchase the product.

Mathematical Perspective

Average interval between consecutive non-zero demand occurrences.

Higher ADI → Less frequent demand.

CV² (Squared Coefficient of Variation)

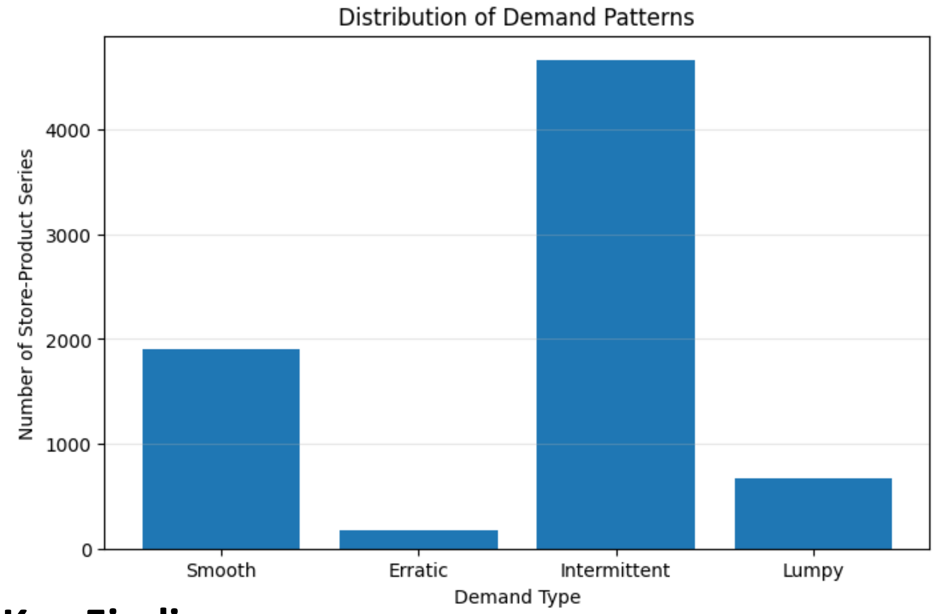
Business Perspective

How consistent the demand quantity is whenever customers purchase.

Mathematical Perspective

Measures variability of non-zero demand.

Higher CV² → More variable demand.



Key Finding

Intermittent demand dominates the dataset (~63%).

Smooth demand is the second largest category.

Erratic and Lumpy demand form a smaller proportion.

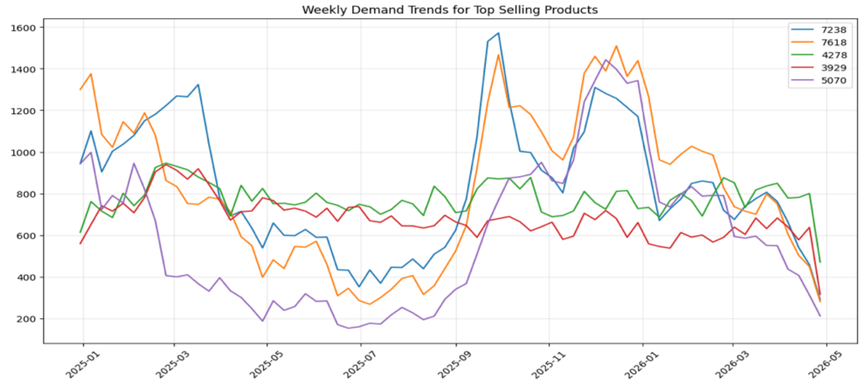
Business Impact

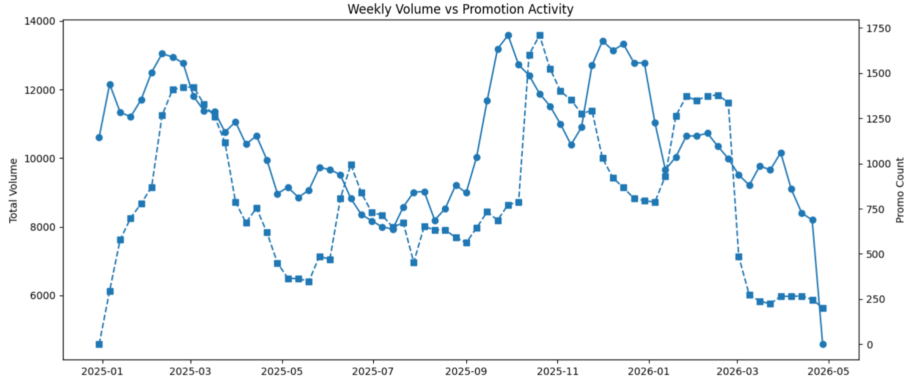
Most products are purchased **infrequently rather than continuously.**

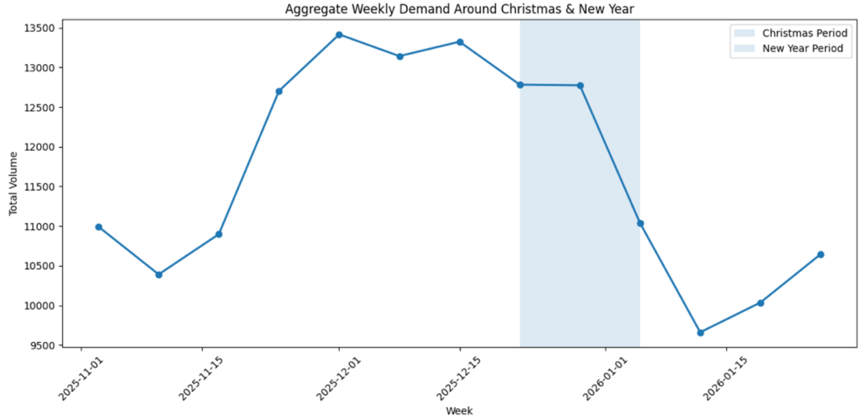
Forecasting Decision

This provided the primary justification for evaluating Croston, SBA and TSB alongside a Moving Average benchmark.

Do Seasonal Events Influence Retail Demand?







Observation 1

Several top-selling products show demand spikes during the same period.

Meaning

Not product-specific. Likely external.

Observation 2

Demand generally increases when promotional activity increases.

However, relationship isn't perfect.

Meaning

Promotions explain part, not all, of demand.

Observation 3

Demand peaks around Christmas and New Year.

Holiday table confirms higher average weekly demand.

Meaning

Calendar events consistently influence customer purchasing behavior.

Key EDA Insights

- Weekly aggregation reduced sparsity.
- Promotions significantly influence demand.
- Demand varies across stores and products.
- Holidays create recurring demand spikes.
- Intermittent demand dominates (~63%).

Forecasting Methodology

Forecasting Pipeline

Raw Daily Sales

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Weekly Aggregation

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Exploratory Data Analysis

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Demand Characterization

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Rolling-Origin Evaluation

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Model Comparison

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Best Forecasting Model

Why Rolling-Origin Evaluation?

- Mimics real-world forecasting
- Uses only historical information
- Prevents data leakage
- Produces realistic performance estimates

EDA guided model selection, while rolling-origin evaluation ensured fair model comparison.

Forecasting Models

Forecasting Models Evaluated

Model	Core Idea	Suitable For
Moving Average	Average recent demand	Stable short-term patterns
Croston	Separates demand size & interval	Intermittent demand
SBA	Bias-corrected Croston	Intermittent demand
TSB	Demand probability + demand size	Highly intermittent demand

Why These Models?

- Traditional baseline
- Specialized intermittent forecasting methods
- Direct comparison under identical evaluation

The objective was not to prove one model superior, but to determine which model best fits this dataset.

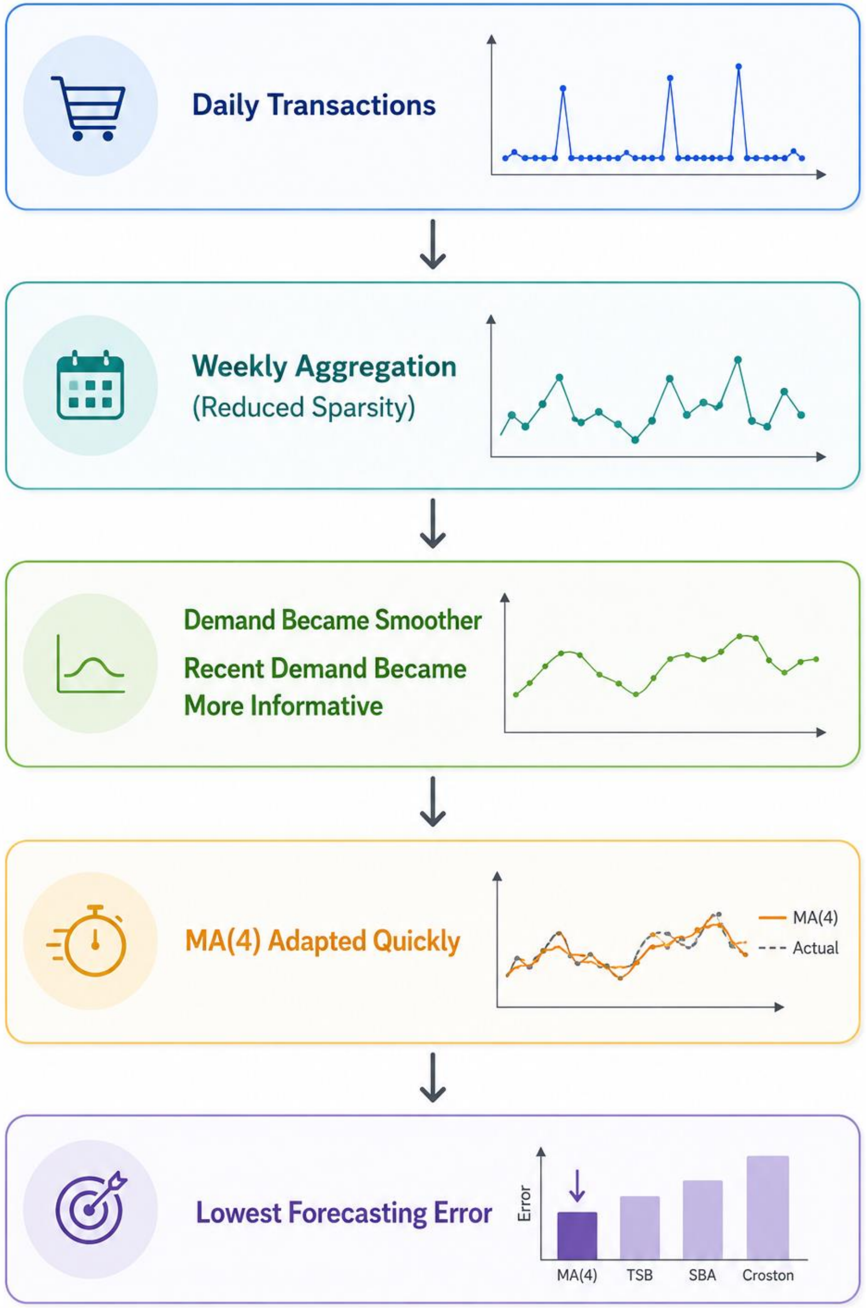
Model Evaluation Strategy

- Evaluation Method
  - 🔄 **Rolling-Origin Evaluation**
    - Forecast one week ahead
    - Uses only historical information
    - Repeated across the forecasting horizon
    - Mimics real-world forecasting

- **Evaluation Metrics**
  - 📉 **MAE (Mean Absolute Error)**
    - Average forecasting error.
    - Easy to interpret.
  - 📈 **RMSE (Root Mean Squared Error)**
    - Penalizes larger forecasting errors more heavily.
    - Useful because large inventory errors are more costly.
  - ⚖️ **Relative Bias**
    - Measures systematic over- or under-forecasting.
    - Closer to zero is better.
  - ⏱️ **Runtime**
    - Measures computational efficiency.
    - Important when forecasting thousands of store-product combinations.

Forecasting Results

Model	MAE	RMSE	Relative Bias	Runtime
MA(4)	**0.943**	**1.546**	**0.066**	**0.14 s**
TSB	0.981	1.548	0.152	6.35 s
SBA	1.138	1.699	0.204	5.99 s
Croston	1.168	1.738	0.267	6.86 s



Why Did MA(4) Outperform Specialized Intermittent Methods?

- **Why not Croston / SBA / TSB?**
  - Designed for **highly intermittent** demand.
  - Estimate demand occurrence as well as demand size.
  - After weekly aggregation, the additional modelling complexity provided **limited benefit**.

Weekly aggregation fundamentally changed the forecasting problem. Once sparsity was reduced, recent demand became a stronger predictor than explicitly modelling demand intervals.

Business Recommendations

- Deploy MA(4) as the baseline forecasting model for weekly store-product demand.
- Incorporate promotion and holiday features into future ML forecasting models.
- Continue forecasting at the **store–product–week** granularity.
- Explore machine learning models (e.g., XGBoost/LightGBM) using the engineered features as future work.

Acknowledgements

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Thank You, Questions?